



NIKHIL PATHAK

AI & Full-Stack Software Engineer

B.Tech. Electronics & Communication Engineering (2028)

National Institute Of Technology, Surat

CGPA: 9.2/10.0

+91-9455281616

nikhilpathakgonda@gmail.com

GitHub

PROFESSIONAL SUMMARY

Software engineer who ships AI systems end to end — Python, SQL, FastAPI, React, Docker. Built a bill-of-materials engine for a bulk-bag manufacturer, calibrated against 441 real work orders, plus agentic LLM workflows with human approval gates and a knowledge-graph document copilot. Paper accepted at the CVPR NTIRE 2026 workshop.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (ECE)	National Institute of Technology, Surat	9.2/10.0	2024-2028
Senior Secondary (XII)	Narayan Public School, CBSE	88.0%	2023-2024
Secondary (X)	[FILL: School Name, Board]	[FILL: %]	2021-2022

TECHNICAL SKILLS

- Languages:** Python, C/C++, SQL, JavaScript, HTML/CSS
- CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, OS, System Design
- Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Redis, Neo4j
- Backend & Web:** FastAPI, REST APIs, React, Next.js, Streamlit, Docker, Git/GitHub
- AI/ML:** PyTorch, Hugging Face, LangGraph, RAG & GraphRAG, LoRA/QLoRA, Pandas

KEY PROJECTS

- AI Bill-of-Materials Engine — Manufacturing & Procurement Automation** 2026
Rules engine, retrieval and constrained LLM, built for a live FIBC bulk-bag manufacturer GitHub
 - Converts a plain-English customer enquiry into an auditable BOM: a deterministic rule engine, hybrid retrieval over 441 historical work orders, and a constrained LLM synthesis step.
 - Gated every output behind 11 validation checks that recompute all arithmetic — median weight error 8.9%, component precision 0.94 / recall 0.96, with an audit trail per run.
- Industrial Knowledge Graph & Document-Intelligence Copilot** Jul. 2026
ET AI Hackathon 2026 — engineering drawings and maintenance records to a queryable graph GitHub
 - Converts P&ID drawings and maintenance/SOP documents into a Neo4j knowledge graph via tiling, fine-tuned YOLO symbol detection, OCR association and cross-tile graph assembly.
 - A QLoRA fine-tuned model translates questions into Cypher, fused with vector retrieval to return cited answers; served as a Dockerised FastAPI service.
- Surat Traffic Intelligence — Real-Time Analytics & Decision Support** Jul. 2026
E-Rakshak Hackathon 2026 (PS-08) — seven live CCTV feeds to an operator dashboard GitHub
 - Per-lane density, adaptive signal optimisation and bottleneck detection from live feeds; signal re-timing cut modelled junction delay by 32.7%.
 - Detection recall collapses under gridlock (463 vs 7,664 detections), so I added a texture-based occupancy fallback to keep site ranking correct. Streamlit dashboard with forecasting.
- LinkMind — Multi-Agent Workflow with Human-in-the-Loop Approval** Jun. 2026 - Present
LangGraph orchestration over FastAPI and PostgreSQL GitHub
 - Two-tier agent stack with a mandatory human approval gate before any action executes, backed by PostgreSQL so long runs pause, resume and stay auditable.
- MELD-SR: Frequency-Guided Super-Resolution (CVPR 2026)** Jan. 2026 - Present
Published multi-expert fusion model for single-image super-resolution GitHub
 - Authored a paper accepted at the CVPR NTIRE 2026 workshop; ranked 19th and 17th globally, and topped the board through the evaluation phase.
- Other Shipped Projects** 2025 - 2026
Full-stack platforms and agentic AI systems GitHub
 - ECO-ROUTE AI routing platform (Next.js, FastAPI, Redis, Docker); multilingual WhatsApp sales agent with deterministic order pricing; LangGraph email-outreach agent; Stonic 2.0 voice desktop agent.

ACHIEVEMENTS

- CVPR NTIRE 2026 – Authored Accepted Paper** 19th & 17th globally; 1st through the evaluation phase 2026
- Academic** 9.2/10 CGPA at NIT Surat, near the top of the batch 2024-Present
- Hackathons** Built complete working systems for ET AI Hackathon (PS-8) and E-Rakshak (PS-08) 2026